

Adaptive Learning App for ADHD in Lima Schools

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Abstract. Education is fundamental for all children, but those with neurological disorders, such as attention deficit hyperactivity disorder (ADHD), have specific learning needs that are not addressed by traditional educational methods. This project focuses on the creation of a mobile application that employs machine learning algorithms to evaluate the progress of these children through regression methods, focusing on mathematical problem solving in elementary schools in Lima. Results have shown that this initiative has improved the understanding and performance of children with ADHD, highlighting the potential of mobile technology and machine learning to move towards a more inclusive and effective education for this group of students. This project has proven to be a valuable tool in the educational field. The inclusion of technological tools such as the mobile app has allowed personalization in teaching, adapting the content and pace of learning to the individual needs of each student with ADHD. In addition, the use of machine learning has enabled a continuous and accurate monitoring of their progress, identifying areas for improvement early on.

Keywords: ADHD · Educational method · children · student · mindfulness · random forest · machine learning · Lima

1 Introduction

Facing the challenges of attention deficit hyperactivity disorder (ADHD), an innovative approach based on Machine Learning and the cloud has been presented. This solution focuses on the development of a mobile application that uses regression algorithms, such as Random Forest, to personalize the learning experience by estimating the levels of children with ADHD. This method, divided into four key stages, combines Mindfulness strategies, progressive data capture, analysis through Machine Learning, and result projection. The algorithm's implementation takes place in Azure Machine Learning Studio, leveraging a Linux environment, and achieves an accuracy of 94%. This project aims to enhance the mathematical understanding of children with ADHD, positively impacting their academic performance and quality of life.

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1.1 Context

ADHD is a disorder affecting between 5%–7% of the world's population (Demontis et al. 2018) and affecting people's concentration, focus, ability to follow instructions and organizational skills, which impacts their performance in multiple areas of their lives (Steinau 2013).

It should be noted that ADHD is a genuine neurological disorder, not attributable to lack of effort or discipline, and its origin is linked to a combination of genetic and environmental factors, according to research (NIH, 2021). As a result, a Machine Learning-based solution has been proposed to aid the understanding of people with ADHD by solving mathematical equations. The solution is based on cloud computing, through services such as Azure, to save time and economic resources.

To develop the Machine Learning-based solution, several inputs were needed. These inputs will be the questionnaires of children with ADHD, which contain information about their symptoms and the resolutions of mathematical equations of various levels.

For the development of the mobile application, the Android Studio development environment was used, which is an environment for programming applications for mobile devices. This development environment offers a wide variety of tools, which allows to create attractive and entertaining mobile applications for users.

Solutions have been proposed to improve the learning and skills of children with ADHD through mobile applications. One such solution is the application "Mobile application with reinforcement activities for children with ADHD" (Ruiz-Ledesma, Bautista-Rosales, & Garay-Jiménez, 2022), which offers ADHD activities to improve concentration, and assessments for parents. The use of artificial intelligence-driven personalized assistance tools is also being investigated, according to the article "Artificial intelligence-enabled personalized assistance tools to improve the education of children with neurodevelopmental disorders" (Barua et al., 2022). Also, in Peru, the TCT method, created by Dr. Kazuhiro Tajima Pozo, improves cognitive areas through brief 10-min cognitive training sessions (Opticos Optometristas, n.d.).

2 Contribution

This research focuses on developing a mobile application that provides math exercises to children with ADHD, adapting the difficulty according to their progress through the use of machine learning and regression algorithms. The application uses random forest algorithms to vary the difficulty of the proposed exercises, thus personalizing the learning experience for each child based on their performance and individual characteristics.

As can be seen in Fig. 1, the method consists of 4 steps. The four stages encompassed the design of an effective strategy for addressing ADHD in children, opting for Mindfulness. The subsequent focus on data collection / data acquisition involved developing a mobile app for children to progressively solve tailor-made exercises. The third stage employed machine learning, specifically the Random Forest algorithm, to analyze collected data and personalize treatment based on individual performance. The concluding phase seamlessly integrated the proposed elements, merging Mindfulness techniques, tiered continuous resolution, and machine learning. This synthesis holds

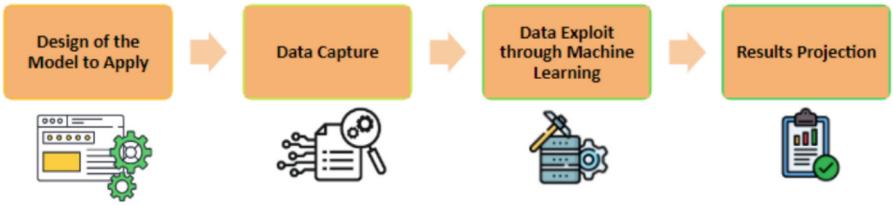


Fig. 1. Steps of the implementation

promising implications for positively impacting the enhancement of ADHD treatment and fostering academic performance improvements in children.

3 Material and Method

3.1 Related Works

The field of mobile applications for children with ADHD, specifically in the context of enhancing mathematical equation-solving skills, has seen considerable research. Unlike previous studies that primarily focus on improving attention, our project concentrates on developing equation-solving skills through relaxation and meditation techniques. Paper 01,"UvaMate: a serious game to learn mathematics for children with ADHD: usability evaluation," assesses the usability of a game designed to aid mathematical learning in children with ADHD (Calleros, Guerrero-García, & Navarro-Rangel, 2020). Teachers express satisfaction with its ease of use but suggest improvements. Paper 02 explores the "Effectiveness of web-based play therapy intervention to support the development of children with ADHD," finding web-based play therapy effective for emotional, behavioral, and social development (Budiyarti, Agustini, Hayati, Panjaitan, & Hakim, 2023). However, it centers more on emotional and social aspects than mathematical understanding. Paper 03 discusses a "Mobile application with reinforcement activities for children with ADHD," focusing on detecting ADHD and improving attention levels through modular exercises and assessments (Ruiz-Ledesma et al., 2022). Paper 04, "Artificial Intelligence-Enabled Personalized Assistance Tools to Enhance the Education of Children with Neurodevelopmental Disorders: A Review," explores AI-based tools for students with neurodevelopmental disorders, emphasizing the need for advanced tools to address current limitations and the importance of personalization in education (Barua et al., 2022).

3.2 Method

The choice of Azure provided a scalable cloud environment, complemented by a Linux setup for computational efficiency. The algorithm processed relevant student data, including academic history and demographic information, as input. Multiple tasks were executed in Azure for testing, resulting in the high accuracy observed. This approach demonstrates the effectiveness of combining cloud-based machine learning tools with a well-configured algorithm for improving predictions in ADHD students' academic outcomes.

The data collection was carried out through the prototype application, where students began solving exercises that were arranged in increasing degree of difficulty. These preliminary data were uploaded to the Azure platform, where the Random Forest algorithm was executed. The Random Forest algorithm was used to determine the level at which the student is performing when solving mathematical exercises, as illustrated in Fig. 2:

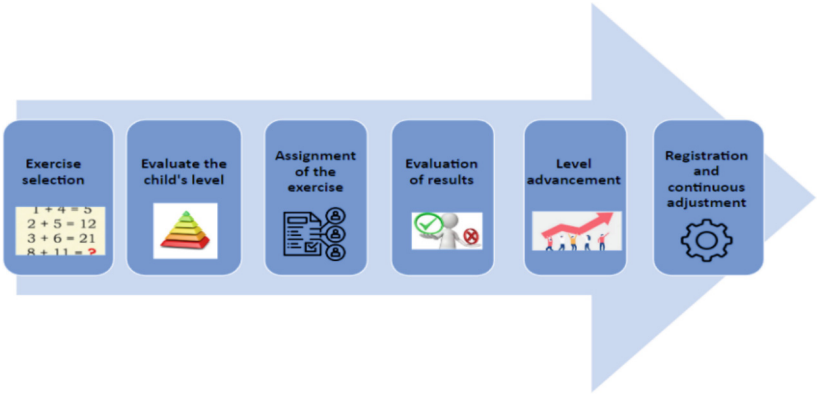


Fig. 2. Process of the random forest algorithm

a. Exercise Selection: When the learner logs into the application, the random forest will be responsible for selecting the next exercise to solve.
b. Evaluate child's level: The random forest will consider the current difficulty level of the student, which was determined based on his/her current grade and previous progress in the application.
c. Assignment of the exercise: Based on the student's level, the application will select a mathematical exercise of equations that is within the expected difficulty.
d. Evaluation of results: The evaluation of results in the application is based on the accuracy of students' answers. If an exercise is solved correctly, a point is added to their progress at that level.
e. Level Advancement: Student progress is based on their performance; solving enough exercises correctly takes them to the next level with higher challenges, while poor performance brings them easier exercises to provide support.
f. Registration and continuous adjustment: The random forest tracks learner performance across exercises and levels, enabling regular adjustments in exercise difficulty and adaptation, ensuring a personalized and effective learning experience.

4 Results

In this section, the results obtained through the experiment designed for improving comprehension and resolution of mathematical problems in children diagnosed with ADHD are discussed. The results show significant advances in the development of mathematical skills, evidencing the effectiveness of personalized approaches in the education of this population.

Experiment Overview: Over a three-week period, 20 children diagnosed with ADHD participated in the experiment. These children aged between 6 to 10 years old, ranging from first through fifth grade of elementary school. This diverse range allowed pedagogical strategies and learning techniques to be adjusted to the individual needs of each child. A crucial approach was to adapt the difficulty of the exercises according to each child's ability to solve mathematical problems. If a child showed progress at a certain level of difficulty, more challenging exercises were presented to encourage growth.

Experiment design: The experiment was conducted in a face-to-face manner, with the participation of the children in a controlled environment. Each child used the mobile application for a period of two weeks, with a frequency of three 30-min sessions per week, the total time of interaction with the application was 120 min each week. During each session, the children solved math exercises, mainly simple addition and subtraction operations, with a gradual increase in difficulty after a certain number of exercises were solved.

Adaptation to Different School Grades: A level system was established, where each level was linked to the corresponding school grade. For example, a child in second grade started at level 0, while a child in fifth grade was at level 3. This adaptation ensured that the exercises were appropriate for each child's level of development and understanding, encouraging consistent progress and avoiding excessive challenges (Rincon-Flores et al., 2023).

Incorporation of Mindfulness Meditation: In the experiment, mindfulness meditation exercises were introduced and performed daily for 10 min at the beginning of each session. These exercises improved the concentration and focus of the ADHD children, as well as maintained them interested in the learning process.

Motivation and Reinforcement: Reward systems were implemented to maintain the children's motivation. After having completed a level of exercises, teachers or parents delivered motivational and congratulatory messages, additionally, upon the successful completion of each session, treats were offered which acted as incentives.

Measurement Variables: The primary variable assessed in the experiment was the math-learning progress of children with ADHD. Pre- and post-experiment tests, along with observations and performance records during the interaction sessions with the app were used to assess this progress.

Privacy Policy: Anonymous information about the use of the app, including system data, was collected. This information was used only to personalize the user experience and improve the functionality of the application. To conduct this experiment, during the process parents were informed about the use of the data and reassured that these would not be shared with third parties to protect security and prevent the unauthorized use of the data. Data collection was conducted solely for academic purposes and not for commercial or production purposes. This transparency and ethical commitment are fundamental in the project's approach, ensuring respect for the privacy and confidentiality of the collected information.

Detailed Results: The results obtained from the experiment are explained below:

a. Experimental protocol: To carry out the experiment, three-level-difficulty adaptive mathematical exercises were designed specifically for children aged 6 to 8 years with ADHD. In the experiment, 20 children participated using a mobile application for a

week, solving a total of 12 exercises per day they had an option of using hints and tips in case help was needed. In addition, meditation breaks and surveys using the Likert scale were implemented to prevent fatigue.

Table 1. Summary of results obtained from 20 students. See Appendix 1 for more details

No	Stage	Iteration	Correct	Incorrect	Current level
1	1	2	24	12	H
2	0	2	26	10	M
3	1	2	20	16	H
4	1	3	19	20	M
5	0	1	22	14	E
6	0	1	19	14	M
7	0	2	21	15	H
8	1	2	20	15	H
9	1	2	22	14	M
10	1	1	24	12	H
11	1	1	24	12	H
12	0	3	22	15	H
13	0	2	19	13	M
14	0	2	21	15	M
15	1	2	23	13	H
16	1	2	22	14	H
17	0	3	24	12	M
18	0	2	21	15	M
19	0	1	18	15	M
20	0	1	24	12	M

b. Results: Despite facing significant challenges while tackling the exercises of greater difficulty, a significant progress in solving medium-difficulty problems was noticed. In the results of the satisfaction surveys, it is important to highlight that the strategy of promoting a positive environment and the inclusion of mindfulness exercises was well-received by the participants (for more details, see Appendix). The project utilized Azure Machine Learning Studio for developing and deploying a Random Forest algorithm, achieving a remarkable 94% accuracy in predicting academic performance for students with ADHD.

Table 1 shows the results in the solution of algebraic problems of a 20-student sample. Some fields included are: stage in which they are, iteration with the highest number of correct answers, number of correct and incorrect answers and the level in which they

are in the last iteration (easy, medium or difficult). Overall, the results of the experiment emphasize the importance of customizing educational strategies to the individual needs of children with ADHD.

The physical and logical architecture of the research project is shown below (Figs. 3 and 4):

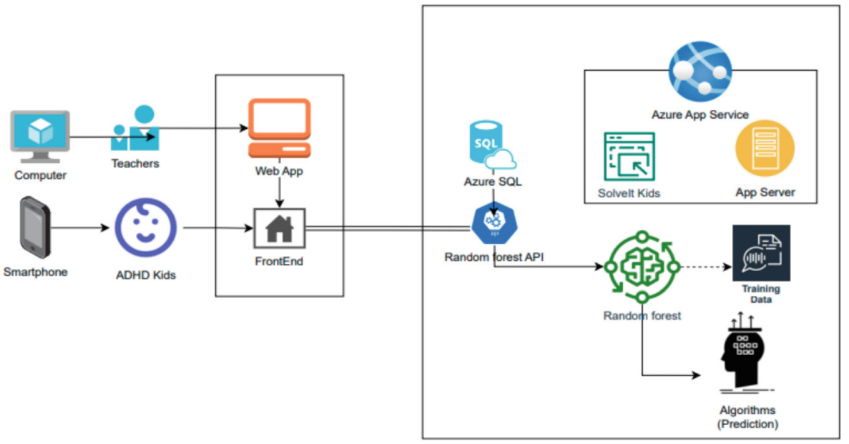


Fig. 3. Physical architecture

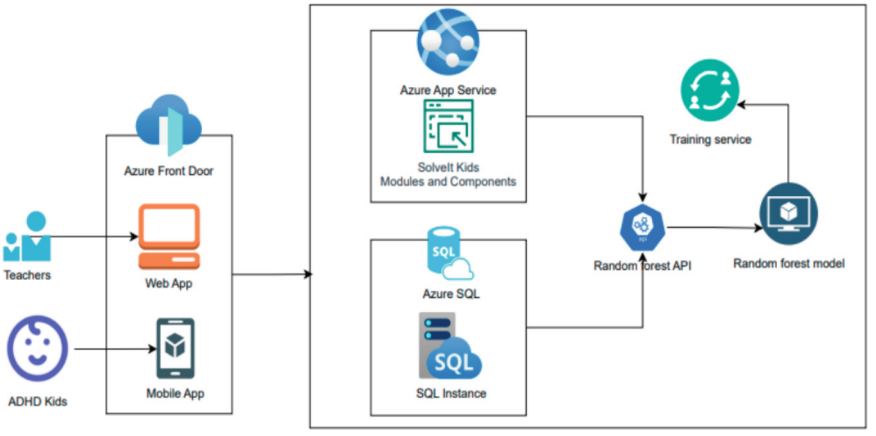


Fig. 4. Logical architecture

In the presented logical and physical architectures, users gain access to the targeted platform via an Android device. This device initiates the front-end interface of the application, prompting children to log in and engage with the exercises. User profiles are managed through a database, while services and the algorithm operate in the Azure cloud, utilizing the Azure database to store student outcomes. Subsequently, the data undergoes processing through the Random Forest algorithm. Caregivers or educators can review the results by connecting through a data analysis program, such as Power BI, equipped with tools to visualize this data through a dedicated web application associated with the program.

5 Conclusions

The method used in the experiment, which involved a mobile application with personalized math exercises for children with ADHD, demonstrated significant progress in the development of mathematical skills. The use of machine-learning and regression algorithms to adapt the difficulty of the exercises according to each child's progress was a key component of the method. Additionally, the incorporation of mindfulness meditation exercises and the implementation of a reward system contributed to maintaining the children's motivation and engagement.

However, it is important to note that the small sample size of 20 children and the short trial duration of two weeks may have limited the generalizability of the results. A larger sample and a longer trial period would possibly provide a more comprehensive understanding of the effectiveness of the method in the long term. Additionally, a longer trial period would allow the observation of sustained progress as well as to identify any potential challenges or limitations that may arise over time.

In conclusion, while the method applied ensures the improvement mathematical skills of children with ADHD, the limitations of the small sample size and short trial duration should be taken into consideration when interpreting the results. Future research with a larger and more random sample, as well as a longer trial period, would provide a more in-depth assessment of the method's effectiveness in the education of children with ADHD.

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Appendix

	Student 1			Student 2			Student 3			Student 4		
Etapa	1			0			1			1		
Iterations	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3
Level	Easy	Easy	Hard	Easy	Easy	Hard	Easy	Easy	Hard	Easy	Hard	Easy
Resolution 1	True	True	True	True	True	False	True	True	True	True	True	True
Resolution 2	True	True	False	True	True	False	True	True	True	True	False	True
Resolution 3	True	True	True	True	False	False	True	True	False	True	False	True
Level	Medium	Medium	Hard	Medium	Medium	Medium	Medium	Medium	Hard	Medium	Medium	Medium
Resolution 4	False	True	False	True	True	True	True	False	False	True	True	True
Resolution 5	True	True	True	False	False	False	False	True	False	True	False	True
Resolution 6	True	True	False	False	False	True	True	True	False	False	True	False
Likert Survey	4	5	3	2	1	4	4	4	3	3	4	5
Level	Hard	Hard	Medium	Easy	Easy	Hard	Hard	Hard	Medium	Hard	Hard	Hard
Resolution 7	False	False	True	True	True	False	False	True	True	False	False	True
Resolution 8	False	False	True	False	True	False	False	False	False	False	False	False

(*continued*)

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	Student 1			Student 2			Student 3			Student 4		
Etapa	1			0			1			1		
Iterations	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3
Resolution 9	False	True	True	True	True	True	False	False	True	False	False	False
Level	Medium	Medium	Hard	Medium	Medium	Medium	Medium	Medium	Hard	Medium	Medium	Medium
Resolution 10	False	True	True	True	True	True	True	True	False	True	False	True
Resolution 11	False	True	True	False	True	True	False	True	True	False	False	True
Resolution 12	True	False	False	False	True	True	False	True	False	False	True	False
Likert Survey	3	4	3	4	3	5	3	4	5	4	5	3
Current Level	Medium	Medium	Hard	Medium	Medium	Medium	Medium	Medium	Hard	Medium	Medium	Medium
Total Easy Exercises Solved Correctly	3	3	0	5	5	0	3	3	0	3	0	3
Total Easy Exercises Solved Incorrectly	0	0	0	1	1	0	0	0	0	0	0	0

(*continued*)

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	Student 1			Student 2			Student 3			Student 4		
Etapa	1			0			1			1		
Iterations	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3
Total Medium Exercises Solved Correctly	3	5	3	2	4	5	3	5	2	3	3	4
Total Medium Exercises Solved Incorrectly	3	1	0	4	2	1	3	1	1	3	3	2
Total Hard Exercises Solved Correctly	0	2	5	0	0	5	0	1	3	0	2	1
Total Hard Exercises Solved Incorrectly	3	1	4	0	0	1	3	2	6	3	7	2

	Student 1			Student 2			Student 3			Student 4		
Etapa	1			0			1			1		
Iterations	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3
Level	Easy	Easy	Hard	Easy	Easy	Hard	Easy	Easy	Hard	Easy	Hard	Easy
Resolution 1	True	True	True	True	True	False	True	True	True	True	True	True
Resolution 2	True	True	False	True	True	False	True	True	True	True	False	True
Resolution 3	True	True	True	True	False	False	True	True	False	True	False	True
Level	Medium	Medium	Hard	Medium	Medium	Medium	Medium	Medium	Hard	Medium	Medium	Medium
Resolution 4	False	True	False	True	True	True	True	False	False	True	True	True
Resolution 5	True	True	True	False	False	False	False	True	False	True	False	True
Resolution 6	True	True	False	False	False	True	True	True	False	False	True	False
Likert Survey	4	5	3	2	1	4	4	4	3	3	4	5
Level	Hard	Hard	Medium	Easy	Easy	Hard	Hard	Hard	Medium	Hard	Hard	Hard
Resolution 7	False	False	True	True	True	False	False	True	True	False	False	True
Resolution 8	False	False	True	False	True	False	False	False	False	False	False	False

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(*continued*)

	Student 1			Student 2			Student 3			Student 4		
Etapa	1			0			1			1		
Iterations	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3
Resolution 9	False	True	True	True	True	True	False	False	True	False	False	False
Level	Medium	Medium	Hard	Medium	Medium	Medium	Medium	Medium	Hard	Medium	Medium	Medium
Resolution 10	False	True	True	True	True	True	True	True	False	True	False	True
Resolution 11	False	True	True	False	True	True	False	True	True	False	False	True
Resolution 12	True	False	False	False	True	True	False	True	False	False	True	False
Likert Survey	3	4	3	4	3	5	3	4	5	4	5	3
Current Level	Medium	Medium	Hard	Medium	Medium	Medium	Medium	Medium	Hard	Medium	Medium	Medium
Total Easy Exercises Solved Correctly	3	3	0	5	5	0	3	3	0	3	0	3
Total Easy Exercises Solved Incorrectly	0	0	0	1	1	0	0	0	0	0	0	0

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	Student 1			Student 2			Student 3			Student 4		
Etapa	1			0			1			1		
Iterations	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3
Total Medium Exercises Solved Correctly	3	5	3	2	4	5	3	5	2	3	3	4
Total Medium Exercises Solved Incorrectly	3	1	0	4	2	1	3	1	1	3	3	2
Total Hard Exercises Solved Correctly	0	2	5	0	0	5	0	1	3	0	2	1
Total Hard Exercises Solved Incorrectly	3	1	4	0	0	1	3	2	6	3	7	2

	Student 9			Student 10			Student 11			Student 12		
Etapa	0			0			0			1		
Iterations	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3
Level	Easy	Easy	Hard	Easy	Easy	Hard	Easy	Easy	Hard	Easy	Easy	Easy
Resolution 1	True	True	False	True	True	True	True	True	True	True	True	True
Resolution 2	True	True	True	True	True	True	True	True	True	True	True	True
Resolution 3	True	True	False	True	True	True	True	True	False	True	True	True
Level	Medium	Medium	Medium	Medium	Medium	Hard	Medium	Medium	Hard	Medium	Medium	Medium
Resolution 4	False	True	True	False	True	False	False	True	True	True	True	True
Resolution 5	True	False	True	True	True	True	True	False	False	True	True	False
Resolution 6	True	True	False	True	False	False	True	True	True	False	True	True
Likert Survey	4	5	4	4	5	5	3	5	3	4	3	3
Level	Hard	Hard	Hard	Hard	Hard	Medium	Hard	Hard	Medium	Hard	Hard	Hard
Resolution 7	True	False	False	False	True	True	False	False	True	True	False	True
Resolution 8	False	True	False	True	False	True	False	True	True	False	False	True

(*continued*)

(*continued*)

	Student 9			Student 10			Student 11			Student 12		
Etapa	0			0			0			1		
Iterations	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3
Resolution 9	False	False	True	False	False	False	True	False	False	False	False	False
Level	Medium	Medium	Medium	Medium	Medium	Hard	Medium	Medium	Hard	Medium	Medium	Hard
Resolution 10	False	True	False	True	True	True	False	True	True	False	False	False
Resolution 11	False	True	False	False	False	True	False	True	False	True	True	True
Resolution 12	True	True	True	True	True	False	True	False	True	False	False	False
Likert Survey	3	3	4	3	4	5	4	4	3	3	3	4
Current Level	Medium	Medium	Medium	Medium	Medium	Hard	Medium	Medium	Hard	Medium	Medium	Hard
Total Easy Exercises Solved Correctly	3	3	0	3	3	0	3	3	0	3	3	3
Total Easy Exercises Solved Incorrectly	0	0	0	0	0	0	0	0	0	0	0	0

(*continued*)

(*continued*)

	Student 9			Student 10			Student 11			Student 12		
Etapa	0			0			0			1		
Iterations	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3
Total Medium Exercises Solved Correctly	3	5	4	4	4	2	3	4	2	3	4	2
Total Medium Exercises Solved Incorrectly	3	1	2	2	2	1	3	2	1	3	2	1
Total Hard Exercises Solved Correctly	1	1	2	1	1	6	2	1	6	1	0	3
Total Hard Exercises Solved Incorrectly	2	2	4	2	2	3	1	2	3	3	3	3

	Student 13			Student 14			Student 15			Student 16		
Etapa	1			1			0			1		
Iterations	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3
Level	Easy	Easy	Hard	Easy	Easy	Hard	Easy	Easy	Medium	Easy	Easy	Hard
Resolution 1	True	True	False	True	True	False	True	True	True	True	True	True
Resolution 2	True	True	False	True	True	True	True	True	False	True	True	True
Resolution 3	True	True	True	True	True	False	True	True	True	True	True	False
Level	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Hard	Medium	Medium	Hard
Resolution 4	True	True	True	True	True	True	False	True	True	False	False	True
Resolution 5	True	False	True	False	True	True	True	False	False	True	True	False
Resolution 6	False	True	False	True	True	False	True	True	False	True	True	False
Likert Survey	3	3	3	5	5	4	4	5	5	4	5	3
Level	Hard	Hard	Hard	Hard	Hard	Hard	Hard	Hard	Medium	Hard	Hard	Medium
Resolution 7	False	False	True	False	False	False	False	True	True	False	False	True
Resolution 8	True	True	False	True	False	False	False	True	True	False	False	True

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	Student 13			Student 14			Student 15			Student 16		
Etapa	1			1			0			1		
Iterations	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3
Resolution 9	False	False	False	False	False	True	True	False	False	False	True	True
Level	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Hard	Hard	Medium	Medium	Hard
Resolution 10	True	True	False	False	True	True	False	False	True	True	True	True
Resolution 11	False	True	True	True	False	True	True	False	True	False	False	True
Resolution 12	False	False	False	False	True	False	False	True	True	False	True	False
Likert Survey	3	3	5	3	5	3	3	5	5	4	4	4
Current Level	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Hard	Hard	Medium	Medium	Hard
Total Easy Exercises Solved Correctly	3	3	0	3	3	0	3	3	0	3	3	0
Total Easy Exercises Solved Incorrectly	0	0	0	0	0	0	0	0	0	0	0	0

(*continued*)

(*continued*)

	Student 13			Student 14			Student 15			Student 16		
Etapa	1			1			0			1		
Iterations	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3
Total Medium Exercises Solved Correctly	3	4	3	3	5	4	3	2	4	3	4	3
Total Medium Exercises Solved Incorrectly	3	2	2	3	1	2	3	1	2	3	2	0
Total Hard Exercises Solved Correctly	1	1	1	1	0	2	1	3	4	0	1	5
Total Hard Exercises Solved Incorrectly	2	2	2	2	3	4	2	3	2	3	2	4

	Student 17			Student 18			Student 19			Student 20		
Etapa	0			1			1			1		
Iterations	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3
Level	Easy	Easy	Easy	Easy	Easy	Hard	Easy	Hard	Hard	Easy	Easy	Easy
Resolution 1	True	True	True	True	True	False	True	True	False	True	True	True
Resolution 2	True	True	True	True	True	True	True	False	False	True	True	True
Resolution 3	True	True	True	True	True	False	True	False	True	True	True	True
Level	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium
Resolution 4	True	True	True	True	True	True	True	True	True	True	True	True
Resolution 5	False	True	True	True	True	False	True	True	True	True	True	True
Resolution 6	True	False	False	False	False	True	False	False	False	False	True	True
Likert Survey	3	4	4	3	3	3	3	5	4	5	5	3
Level	Hard	Hard	Hard	Hard	Hard	Hard	Hard	Hard	Hard	Hard	Hard	Hard
Resolution 7	False	False	True	False	False	False	False	False	True	False	True	True
Resolution 8	True	True	False	False	True	False	False	False	False	True	False	False

(*continued*)

(*continued*)

	Student 17			Student 18			Student 19			Student 20		
Etapa	0			1			1			1		
Iterations	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3
Resolution 9	False	False	False	True	False	True	False	True	False	False	False	False
Level	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium
Resolution 10	False	False	True	False	True	False	True	True	True	False	True	False
Resolution 11	False	True	True	True	False	True	False	False	False	True	False	True
Resolution 12	True	False	True	False	True	True	True	True	True	False	False	False
Likert Survey	3	3	4	3	4	4	5	5	3	3	4	5
Current Level	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium
Total Easy Exercises Solved Correctly	3	3	3	3	3	0	3	0	0	3	3	3
Total Easy Exercises Solved Incorrectly	0	0	0	0	0	0	0	0	0	0	0	0

(*continued*)

(*continued*)

	Student 17			Student 18			Student 19			Student 20		
Etapa	0			1			1			1		
Iterations	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3	Iteration 1	Iteration 2	Iteration 3
Total Medium Exercises Solved Correctly	3	4	5	3	4	4	4	4	4	3	4	4
Total Medium Exercises Solved Incorrectly	3	2	1	3	2	2	2	2	2	3	2	2
Total Hard Exercises Solved Correctly	1	1	1	1	1	2	0	1	2	2	1	1
Total Hard Exercises Solved Incorrectly	2	2	2	2	2	4	3	2	4	1	2	2

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